

Scalability & Future Plan

Date	02 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan: Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited dataset size and historical records	May reduce prediction accuracy for unseen scenarios	High
2	No real-time weather data integration	Predictions depend on manually entered values	Medium
3	Application is currently deployed locally	Limited accessibility and scalability	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users only	Deploy on IBM Cloud to support multiple concurrent users.
2	Data Storage	Uses a static dataset stored locally	Integrate cloud storage and larger datasets.
3	Performance	Suitable for small-scale prediction tasks	Optimize models and use cloud computing resources.
4	Security	Basic local application security	Implement authentication and secure cloud deployment.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integration of Real-Time Weather APIs	3 – 6 Months	Improves prediction accuracy by automatically collecting live weather data.
Phase 3	Deployment on IBM Cloud and Remote Access	6 – 9 Months	Increases accessibility and allows multiple users to access the system from anywhere.
Phase 4	Automated Flood Alert and Notification System	9 – 12 Months	Provides early warnings to authorities and improves disaster preparedness and response.